

HARARE INSTITUTE OF TECHNOLOGY



DEPARTMENT OF INFORMATION TECHNOLOGY

BTECH (HONS) INFORMATION TECHNOLOGY

**Leveraging Software-Defined Networking (SDN) for Enhancing
Internet Reliability and QoS in Zimbabwe's Broadband Networks**

BY

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**This project report was submitted to Harare Institute of technology in partial fulfilment
of the bachelor of technology (Hons) degree in information technology**

DECLARATION

I, Tanaka Keith Mashoko (H220325F), declare that this project : “ Leveraging Software-Defined Networking (SDN) for enhancing Internet Reliability and QoS in Zimbabwe’s Broadband Networks” is my own original work and has not been submitted to any other institute for the award of any degree or qualification.

Sources provided and used in creation of this document have been duly acknowledged and referenced in accordance with academic standards. Any assistance received during the course of this study has been appropriately recognised.

Signed: _____

Date: _____

DEDICATION

This project is dedicated to my family for the support and encouragement throughout the entire academic journey.

I also dedicate this work to Zimbabwe's evolving IT sector , mostly the telecommunications sector with the hope that the ideas presented in this research contribute meaningfully toward improving national broadband reliability, resilience and digital transformation.

Above all, dedicate this project to every aspiring innovator who believed in me and gave me the courage and motivation to continue even in constrained environments.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my immediate supervisor, Mr. Sumburero, for his invaluable advice , guidance and constructive criticism. The continuous support throughout the duration of this research is much appreciated for his insights and technical direction greatly shaped how the project moved from initial stage to the end of it.

I am also grateful for the support received from the lecturers and staff of the Department of Information of technology (IT) for the academic foundation and support until completion of this study.

ABSTRACT

Reliable broadband connectivity remains a critical driver of socio-economic development, yet Zimbabwe's telecommunications infrastructure continues to face persistent challenges due to economic constraints (ITU 2023). The above issue degrades service continuity and Quality of Service (QoS), while traditional routing protocols such as OSPF and BGP exhibit limited adaptability during dynamic network disruptions.

Zimbabwe Adaptive Network is brought to you by this study , a hybrid Software defined network that is composed of edge-based artificial intelligence for improved network resilience and adaptability. Unlike simulation-based approaches (Lantz, Heller and McKeown 2010), this

implementation leverages real-time traffic generated from distributed edge devices such as Raspberry Pi nodes and ESP32 microcontrollers, which feed live data into Ryu-based SDN controllers.

Addressing computational limitations within the control plane, the framework came up with Heterogeneous SDN Architecture incorporating an external edge inference engine powered by NVIDIA Jetson platforms. The framework integrates (AQOSRM) Adaptive QOS Routing Mechanism which is enhanced by AI-driven insight (Xie et al. 2019).

Experimental validation on a hybrid physical-emulated testbed demonstrates that hybrid provides enhanced traffic prioritization compared to legacy systems , reduced latency and improved fault recovery times. Having the conclusion of SDN with edge AI providing a scalable , cost-effective and intelligent solution for strengthening network resilience in resource-constrained environments such as Zimbabwe.

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List of Acronyms

AQoSRM: Adaptive QoS Routing Mechanism

BGP: Border Gateway Protocol

CapEx: Capital Expenditure

DPID: Datapath Identifier

FX: Foreign Currency

ISP: Internet Service Provider

MPLS: Multi-Protocol Label Switching

MTTR: Mean Time to Repair

OpEx: Operational Expenditure

OSPF: Open Shortest Path First

QoS: Quality of Service

SDN: Software-Defined Networking

TE: Traffic Engineering

ZAN: Zimbabwe Adaptive Network

CHAPTER 1: INTRODUCTION

1.1 Introduction

Having a network to provide connectivity between individuals also comes along with ensuring high levels of resilience, adaptability and quality of service (QoS). However in developing countries , Zimbabwe , there are several factors that can affect or impact service continuity such as limited access to foreign currency for infrastructure upgrades and reliance on aging legacy , the mentioned above can then impact service continuity, leading to increased latency, packet loss, and prolonged network outages.

Traditional networking paradigms, based on distributed routing protocols such as OSPF and BGP, are inherently reactive and lack the adaptability required by the service to respond effectively to dynamic and non-localized disruptions. These low speeds support widespread failures resulting in high Mean Time to Repair (MTTR), making them unsuitable for environments characterized by systemic instability.

ZAN (Zimbabwe Adaptive Network) is then proposed as the cure to the existing problem , having the ZAN as the advanced hybrid Software-Defined Networking (SDN) architecture augmented with edge-based artificial intelligence. By decoupling the plane from the data plane and including the edge layer, the framework enables centralized control , real-time analytics and predictive decision-making.

The proposed system will have edge devices such as ESP32 microcontrollers and Raspberry Pi nodes to collect real time network telemetry. The collected data is processed and assessed by an external AI inference engine powered by NVIDIA Jetson platforms, this will allow the creation of an edge-inference integration layer. This layer now will enhance the SDN controller's capabilities by enabling intelligent traffic analysis, adaptive routing decisions and anomaly detection without overloading the control plane.

Through this combination of edge intelligence and SDN programmability , framework strive to deliver a scalable , resilient networking solution.

1.2 Motivation

The performance of Zimbabwe's telecommunication infrastructure is facing significant challenges such as system fragility caused by disrupted network operations and compromised service continuity. Having to change the whole infrastructure is not possible currently with the issue of limited foreign currency available in the ISPs pushing forward the need of the hybrid systems that can substitute cost with reliability and availability resources. Despite growing internet penetration , end users have poor Quality of Service Qo , that is high latency , inefficient routing and packet loss which affects real-time applications such as telemedicine, VoIP and online learning.

1.3 Problem Statement

The Traditional IP networking paradigms such as OSPF,BGP and MPLS find it hard to maintain services continuity and acceptable Quality of service (QoS) in environments affected by systemic power instability and economic constraints. Moreover , these legacy systems lack real-time network user communication, predictive fault detection, intelligent traffic prioritization, cost-effective scalability. Therefore, a hybrid system is of need that will come with edge-based artificial intelligence and a networking framework that integrates SDN with the AI to enable adaptive and real-time network management.

1.4 Related Work

Massive research has been done in the area of Software-Defined Network , Quality of Service and network resilience. SDN has proven to improve network performance by centralized control and global visit making it possible to conduct traffic engineering and dynamic routing optimization.

Advancements also got into artificial intelligence into networking. Machine learning techniques are being used for traffic classification, anomaly detection and predictive routing showing off the improved adaptability and performance .

However, despite all these above verdicts , a significant gap remains. Existing frameworks typically follow on managing and solving these challenges. Moreover, there is limited research on integrating edge-based AI with Software-Defined-Networks in these developing economies, particularly in environments where infrastructure reliability is inconsistent.

To develop the Zimbabwe Adaptive Network (ZAN) , having the hybrid SDN-based framework that combines traditional networking with intelligent, edge-based AI to improve resilience and performance.

To build an Edge-Inference Integration Layer that uses distributed edge devices, powered by NVIDIA Jetson platforms, to process network data in real time and detect anomalies quickly and efficiently.

1.5 Hypothesis

H1 (Quality of Service Hypothesis)

The Adaptive QoS Routing Mechanism (AQoSRM), enhanced with SDN-based traffic engineering, will reduce end-to-end latency and packet loss by over 70% compared to traditional routing approaches.

H2 (Intelligence Hypothesis)

The integration of edge-based AI through the Edge-Inference Integration Layer will improve anomaly detection accuracy and reduce network response time to congestion and failures compared to rule-based SDN systems.

1.6 Technical Objectives

1. To develop an Adaptive Quality-of-Service Routing Mechanism (AQoSRM) for dynamic traffic classification and prioritization of real-time applications.
2. To design and implement the Zimbabwe Adaptive Network (ZAN) framework as a hybrid SDN architecture integrated with edge-based AI.
3. To implement an Edge-Inference Integration Layer with distributed edge devices and NVIDIA Jetson platforms for real-time telemetry processing and anomaly detection.

1.7 Expected Results

Upon a successful implementation of the hybrid these are expected outcomes:

- A validated ZAN framework that integrates SDN and edge AI for resilient network operation.
- A functional Edge-Inference Integration Layer capable of real-time traffic analysis and anomaly detection.
- Reduced MTTR to sub-50 ms during simulated power outage scenarios.
- Significant improvements in latency, packet loss, and overall QoS performance.
- A cost-effective networking solution that minimizes reliance on expensive proprietary hardware.
- A scalable architectural model suitable for deployment in resource-constrained environments.

1.8 Conclusion

This chapter has established the need for an intelligent, resilient and cost-effective networking solution tailored to Zimbabwe's unique operational environment. It highlighted the limitations of traditional networking systems in handling power instability, economic constraints and increasing demand for high-quality digital services.

The proposed Zimbabwe Adaptive Network (ZAN) framework addresses these challenges by combining the programmability of SDN with the intelligence of edge-based AI. By introducing real-time telemetry processing, predictive analytics and power-aware routing, the framework provides a robust foundation for improving network resilience and Quality of Service.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This Chapter looks deeper to the theoretical part of the foundation of the ZAN framework. It consists of the analysis of the existing traditional systems having in mind the Quality of Service (QoS), Software-Defined_networking (SDN , Artificial Intelligence AI as well as network resilience.

The review will be staged, beginning with looking into the core SDN principles and the progress toward advanced concepts such as AI-driven traffic management, monitoring and edge-based processing. Having the now approach critically analyse and evaluate identifying limitations, particularly in the developing economies operating under economical and infrastructural constraints.

2.2 Related Works

2.2.1 Software-Defined Networking Fundamentals

Software-Defined Networking , is a way to move from the traditional way of networking by having architectures that decouple the data plane and the control plane. This gives the SDN a centralized control unit within a logically centralized controller whereas legacy networks routing and decisions are distributed across different devices.

Separation of control plane and data plane comes with :

- Programmability – The network can be controlled and adjusted through software, allowing rapid adaption to changing conditions.
- Simplified management – having the separation of control and data plane , now operational complexity is reduced and enables automation.
- Global network visibility – the controller now maintains a holistic view of the network topology, enabling optimized routing decisions.

Research has stated and demonstrated that SDN improves networking at large having efficient and flexible networks in large scale and dynamic environments. These capabilities make SDN a suitable foundation for addressing the limitations of traditional networks.

2.2.2 QoS Provisioning in SDN Environments

QoS (Quality of Service) is crucial in the modern networking cycle as it is a crucial requirement for latency-sensitive applications such as VOIP, video conferencing and telemedicine.

In SDN, QoS is enhanced through :

- Queue Management : Having to manage which packets go first and arrange them with a defined priority order , packet scheduling , improves bandwidth utilization.
- Application-Aware Routing : Knowing application's requirements and having traffic flows classified and be prioritized on application requirements.
- Dynamic Traffic Engineering (TE) : Having the controller allocate resources based on real-time network conditions

However, most QoS implementations rely on predefined rules and lack adaptive intelligence, limiting their effectiveness in highly dynamic environments.

2.2.3 Network Resilience and Fault Tolerance

Network Resilience being the level of a system to deliver a service or maintain acceptable levels of service in the presence of failures.

This is the major failure part of traditional systems that take several seconds or even minutes due to the routing protocol convergence to recover from failures. This is not ideal for real-time services.

SDN introduces many ways to improve resilience:

- Centralized Recovery Logic : The controller being able to recompute the best path for data flow based on the current global network state.
- Precomputed backup paths: Reduces recovery time during link or node failures.

Despite these advancements, traditional resilience models assume stable infrastructure and do not account for environments where failures are frequent and systemic, such as those caused by power outages.

2.2.4 Edge Computing in Networking

Edge computing has emerged as a transformative paradigm that brings the computation closer to the data sources. Shortening the data transfer and utilizing all the data packets , less if no data loss. This

reduces the data latency as the data is now being processed very closer to the source hence faster responses now. Also improved reliability due to the local processing which ensures continued operation even when connectivity to central systems is disrupted.

In many topologies, especially networking, edge devices such as embedded systems and microcontrollers can collect and also preprocess telemetry data in real time. This is valuable in environments where connectivity is intermittent.

2.2.5 Artificial Intelligence in Networking

The ability of Artificial intelligence, particularly machine learning (ML), to automate, analyse and predict the state of an incident before the instance has made it so useful to the smooth flow of networking.

Key applications of Artificial intelligence to networking include:

- Anomaly Detection – having to monitor and be ready to capture unusual patterns or unusual network behaviour or even security threats.
- Traffic prediction – Monitoring current state of network load to have forecast network load so as to enable proactive resource allocation.
- Adaptive Routing – dynamically adjusting routes based on learned network conditions.

AI-driven systems have the advantage over traditional systems because they learn quickly from historical data and they adapt to changing environments.

However, most AI implementations rely on centralized processing, which introduces challenges such as :

- High latency
- Scalability limitations

2.2.6 Integration of AI with SDN

Integration of SDN and AI has become the best solution to enhance network reliability. In AI-enabled architectures :

- The SDN controller provide the centralized control and network visibility.

- AI model analyses network , monitor data traffic and predict and generate network insights.
- The controller gets the insights and optimize routing and resource allocation.

The combination of these two brings a powerful :

- Predictive network management
- Automated decisions
- Improved adaptation of dynamic conditions.

Despite its potential, having AI integrated directly into the SDN controller can lead to performance bottlenecks as controllers are not optimized for intensive data processing.

2.2.7 Edge AI for Intelligent Networking

Edge AI , takes advantage of the edge—computing and does the analysis on the edge devices that are available and also the nearby processing units. In networking this approach involves collecting telemetry data at edge devices, processing data using AI models locally and also sending actionable insights to the SDN controller.

Advantages of edge AI include:

- Real-time analysis is done at the edge devices hence no distant servers are used leading to low latency decision making.
- Having the AI do more work and offloading the load on the SDN Controller.
- Resilience, that is having to continue operations during network disruptions.

Platforms such as NVIDIA Jetson enable efficient deployment of AI models at the edge, providing sufficient computational power for real-time inference. Edge Artificial intelligence is suitable for environments with limited infrastructure reliability, as it enables localized intelligence and reduces dependence on centralized systems.

2.2.8 SDN in Developing Economies

Research on SDN in developing countries proved to have come up with challenges:

- High Cost of Infrastructure Upgrades
- Dependence on Legacy Systems

➤ Limited Technical Expertise

Most networking systems when under migrations or upgrades take the SDN architectures , which combine the SDN-enabled devices with the legacy infrastructure.

However, most existing studies focus on scalability and cost with limited attention to real-time intelligence.

2.3 Research Gap

The literature reviews that while significant progress has been made in SDN and Quality of Service optimization and AI-driven networking, the existing solutions fail to address the combined challenges present in developing economies. Specifically there is lack of a system that combines the SDN-architecture and AI for real-time analytics.

Existing approaches typically address the following in isolation:

- ❖ Having SDN to improve control and programmability
- ❖ AI enhanced intelligence and adaptability

However the needed framework is a framework that combine these technologies to operate under the conditions of economic constraints.

2.4 Conclusion

This chapter has reviewed the foundational concepts and recent advancements of Quality of Service , Artificial intelligence, network resilience and also AI in networking. The analysis confirms that AI can be a strong analytic and network forecasting tool, while SDN can be best in having the data plane and control plane decoupled and edge computing enabling low-latency. However, existing frameworks do not fully address the unique constraint environment of developing economies such as Zimbabwe.

This identified gap motivates the development of the ZAN (Zimbabwe Adaptive Network) framework, which integrates hybrid SDN with edge AI to deliver a resilient, intelligent and cost-effective networking solution tailored for volatile infrastructural landscapes.

CHAPTER 3: REQUIREMENTS ANALYSIS

3.1 Introduction

Having the ZAN framework , it increases the network performance , maintenance and also resilience. It will have an integration of artificial intelligence and SDN in a way to work under all the constraints of Zimbabwe's environment, characterized by economic limitations. The chapter will take a step into the requirements needed by the system to be functional so as to deliver the service.

3.1.1 Evaluation of Alternatives

Three strategies were evaluated to address the challenges of high latency, packet loss and slow convergence :

a. Maintain Legacy OSPF/BGP Infrastructure

Analysis:

Traditionally routing uses distributed decision-making. During widespread failures such as load shedding, convergence times are slow, resulting in prolonged service disruptions and high Mean Time to Repair (MTTR).

Limitations:

- No centralized control
- No real-time intelligence
- No adaptation and prediction of failures

Verdict: Rejected

This approach can not meet the resilience and QoS requirements of modern networks.

b. Deploy Proprietary Solutions (MPLS-Based Networks)

Analysis:

Multiprotocol Label Switching (MPLS) provides strong QoS guarantees and reliable performance. However, it requires expensive proprietary hardware and significant foreign currency investment.

Limitations:

- High Capital Expenditure (CapEx)

- Vendor lock-in
- Scalability and flexibility is very limited

Verdict: Rejected

This approach is economically feasible in constrained environment

c. Develop Software-Defined Network with integrated artificial intelligence

Analysis:

A hybrid SDN architecture allows the integration of artificial intelligence (AI) with SDN-enabled devices. Artificial intelligence allows the system to have predictive capabilities and real-time network analysis.

Advantages:

- Having distributed intelligence with a centralized control
- Cost-effective as we use existing infrastructure
- Real-time network analyses and anomaly detection
- Also real-time adaptive routing

Verdict: Accepted

The above stated framework satisfies both technical and economic constraints.

3.1.2 Development Strategy

The Zimbabwe Adaptive Network is has three-layer architecture

1. Data Plane

- OpenFlow-enabled switches
- Legacy IPv4 routers

2. Control Plane

- SDN controller (Ryu/ONOS)
- Applying the routing logic and network policies.

3. Intelligence Layer (AI)

- Raspberry Pi and ESP32 devices for telemetry collection.
- NVIDIA Jetson platform for AI-based assistance.

Edge-Inference Integration Layer

This layer is responsible for the communication of the SDN devices and artificial intelligence, enabling

- Real-time telemetry data streaming
- AI-driven anomaly pattern detection
- Adaptive routing decisions

3.2 User Requirements

Primary users of the system are:

- ❖ Internet Service Providers (ISPs)

Users require the system so they can continue having reliable connectivity while providing real-time automation on the network.

3.2.1 Requirements Collection

The requirements of this system were identified through the contextual analysis of the Zimbabwe's telecommunications environment.

1. QoS Requirement

The system must ensure high-quality of service for all critical paths

- Reduce latency and packet loss
- Maintain jitter below 30ms for real-time applications.

2. Intelligence Requirement

The system will have intelligence of detecting and monitoring traffic trends

- Detect anomalies in network traffic
- Predict potential failures or congestion
- Adapt routing decisions dynamically

4. Scalability Requirement

In terms of scalability , the system must support incremental expansion:

- Adding new nodes in the network without network failure or disruption
- Support of hybrid integration of legacy and SDN devices

3.2.2 Technical Feasibility

This proposed system is technically feasible due to the availability of tools and widely supported technologies :

- SDN Protocols: OpenFlow
- Controllers: Ryu / ONOS
- Programming Language: Python
- AI Frameworks: TensorFlow, PyTorch, Scikit-learn

3.2.3 Hardware Requirements

The system was designed to operate with cost-effective and accessible hardware components :

Core Components

- Controller Server: Hosts SDN controller and control logic
- OpenFlow Switches: Enable SDN functionality
- Legacy Routers: Maintain backward compatibility

Edge Layer Components

- Raspberry Pi Nodes: Collect and preprocess network telemetry
- ESP32 Microcontrollers: Lightweight traffic monitoring and sensing
- NVIDIA Jetson Platform: Executes AI models for real-time inference

This combination ensures a balance between performance and cost -efficiency.

3.2.4 Software Requirements

Networking Tools

- Ryu / ONOS (SDN controller)

- iPerf (traffic generation)

Programming and Integration

- Python (core implementation)
- REST APIs / MQTT

These tools will help to make all the data collected and processed give useful predictive analyses .

3.2.5 Technical Expertise

For a successful implementation one should have expertise in:

- Networking protocols (OSPF, BGP, MPLS, OpenFlow)
- SDN architecture and controller development
- Python programming
- Machine learning fundamentals
- Data streaming and telemetry processing

3.3 Economic Feasibility

3.3.1 Cost-Benefit Analysis

Hybrid gives significant economic advantages:

- Cost Savings
- Reuse of legacy infrastructure reduces CapEx
- Open-source tools eliminate licensing costs
- Edge devices (Raspberry Pi, ESP32) are low-cost

Operational Efficiency

- Automated fault recovery reduces OpEx
- AI-driven optimization minimizes resource wastage

3.3.2 Tangible Benefits

- Reduces infrastructure upgrade costs
- Improved network uptime and network

3.3.3 Intangible Benefits

- Improved services , crucial digital services such as e-health, e-learning
- Increased reliability of financial and government systems
- improved user trust and satisfaction

3.4 Operational Feasibility

Intelligent Operations

Artificial intelligence had to enable:

- Real-time anomaly detection
- Adaptive traffic management
- Reduced dependency on manual intervention
- Regulatory Compliance

Centralized control enables:

- Traffic prioritization for essential services
- Policy enforcement aligned with national regulations

3.5 Work Plan

3.5.1 Work Schedule

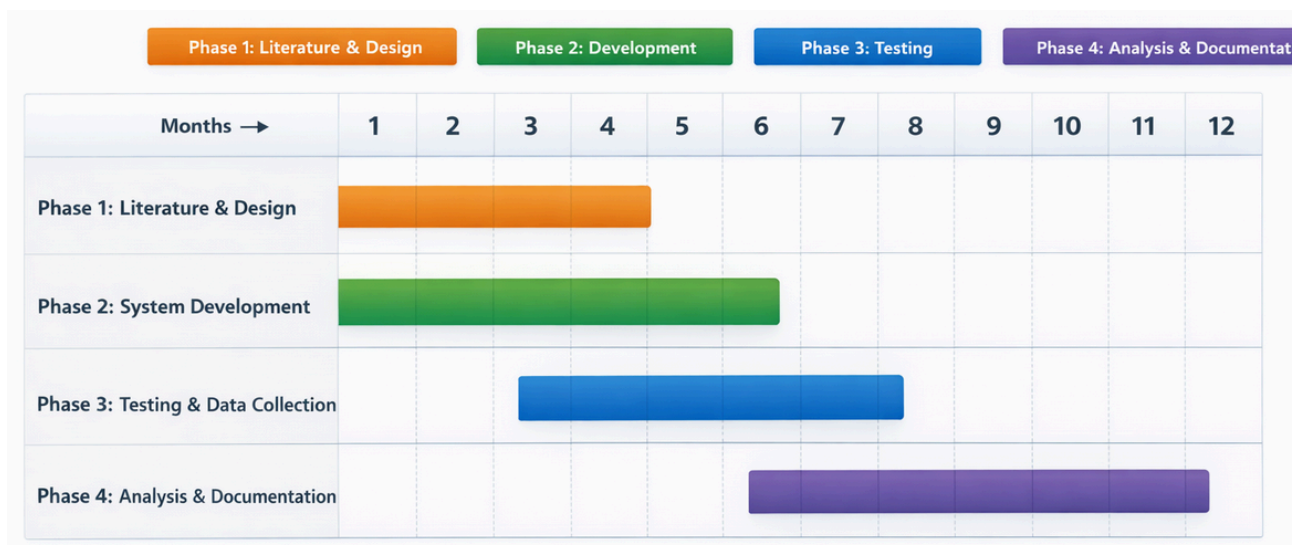
Table 1.0

Phase	Duration	Activities
Phase 1: Literature & Design	Months 1–3	Architecture design, PRC-CPP formulation
Phase 2: System Development	Months 4–6	SDN setup, edge device integration, AI model development

Phase 3: Testing & Data Collection	Months 7–9	telemetry analysis
Phase 4: Analysis & Documentation	Months 10–12	Performance evaluation, thesis writing

3.5.2 Gantt Chart Description

Fig 1.0



3.6 Conclusion

This chapter has defined the comprehensive requirements for implementing the Zimbabwe Adaptive Network (ZAN) framework. The analysis confirms that a hybrid SDN and Edge AI approach is both technically and economically viable, as edge intelligence enhances real-time decision-making and reduces control plane load. The system can operate effectively under Zimbabwe’s power and economic constraints, utilizing open-source tools and low-cost hardware to make implementation feasible. Overall, the requirements analysis validates the practicality of

deploying a resilient, intelligent and scalable network architecture capable of significantly improving QoS and service continuity.

CHAPTER 4: SYSTEM ANALYSIS

4.1 Introduction

The chapter will look into the analysis of both the existing traditional legacy systems and also the proposed hybrid system ZAN framework. The Proposed system now will work with the edge-devices to collect information or data and uses its privilege of artificial intelligent integration to come up with the best functional architecture .

Unlike traditional systems that look solely on data and control plane behaviour , this chapter now shows how the ZAN incorporates an additional intelligence layer that will be pulling real-time telemetry processing and AI-driven decision-making.

4.2 Description of the Current System

The existing network infrastructure used by Internet Service Providers (ISPs) in Zimbabwe is based on traditional IP networking principles.

Key Characteristics

- Distributed Control Plane:

Routing decisions are made independently by each router using protocols such as OSPF and BGP.

- Tightly Coupled Architecture:

Control and data planes reside within the same physical devices.

- Static Configuration:

Network policies are manually configured and difficult to modify dynamically.

- Hardware Dependency:

Reliance on proprietary vendor equipment (e.g Cisco, Juniper).

4.2.1 Limitations of the Current System

The analysis reveals several critical weaknesses:

1. High MTTR (Slow Failure Recovery)

- Routing protocols require convergence after failures.
- Recovery may take seconds to minutes.

2. Lack of QoS Awareness

- No fine-grained traffic prioritization.

3. No Real-Time Intelligence

- No telemetry-driven monitoring.
- No unique patterns detected and no ability to predict.

4. Power-Unaware Routing

- Re-routing if there is node failure

5. Economic Inefficiency

- Migration requires expensive infrastructure

4.3 Overview of the Proposed System (ZAN Framework

The ZAN , Zimbabwe Adaptive Network aims to improve legacy systems by introducing the use of artificial intelligence in the network to analyse, monitor and have a predictive network forecast.

4.3.1 ZAN System Architecture

Layers of the system :

1. Data Plane (Infrastructure Layer)

- OpenFlow-enabled switches
- Legacy IPv4 routers
- It is responsible for packet forwarding

2. Control Plane (SDN Controller Layer)

- Ryu SDN Controller
- Executes routing decisions

Implements: AQoS RM (QoS routing)

3. Intelligence Layer (Edge AI Layer)

Components:

- Raspberry Pi - telemetry collectors)
- ESP32 microcontrollers (lightweight sensing)

4. Edge-Inference Integration Layer

Acts as a bridge between:

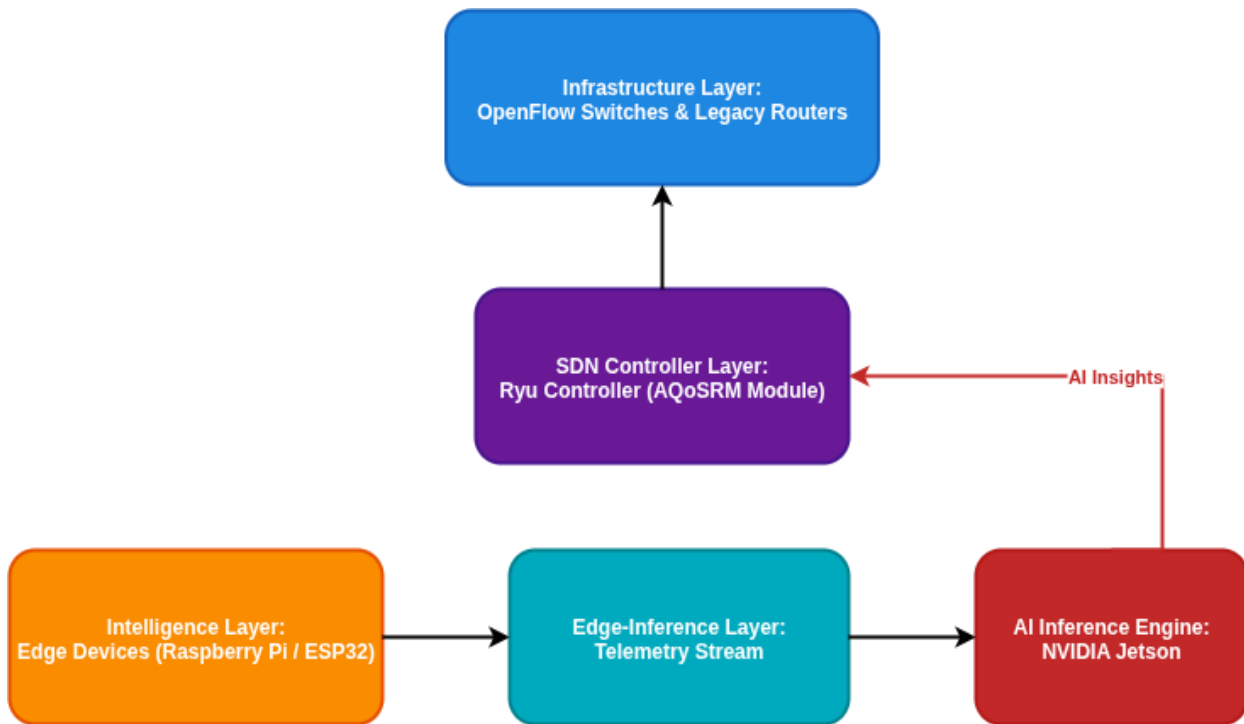
- Intelligence Layer
- SDN Controller

Functions:

- Streams telemetry data
- Performs AI inference
- Sends insights to controller

4.3.2 System Architecture Diagram

Fig 2.0



4.4 Functional Analysis of the Proposed System

4.4.1 Functional Requirements

System will bring the following functionalities:

1. Topology Discovery

- The system should be able to detect the nodes available
- Maintain the topology state to apply routings

2. Real-Time Telemetry Collection

- Edge devices collect:
 - Traffic volume
 - Latency
 - Packet loss
- Data streamed to AI engine

3. AI-Based Anomaly Detection

- AI in support on anomaly detection and alerting on unusual patterns detected

4. Adaptive QoS Routing (AQoSRM)

- Classification of network and bandwidth prioritization.

7. Dynamic Flow Rule Installation

- Ensures optimal routing

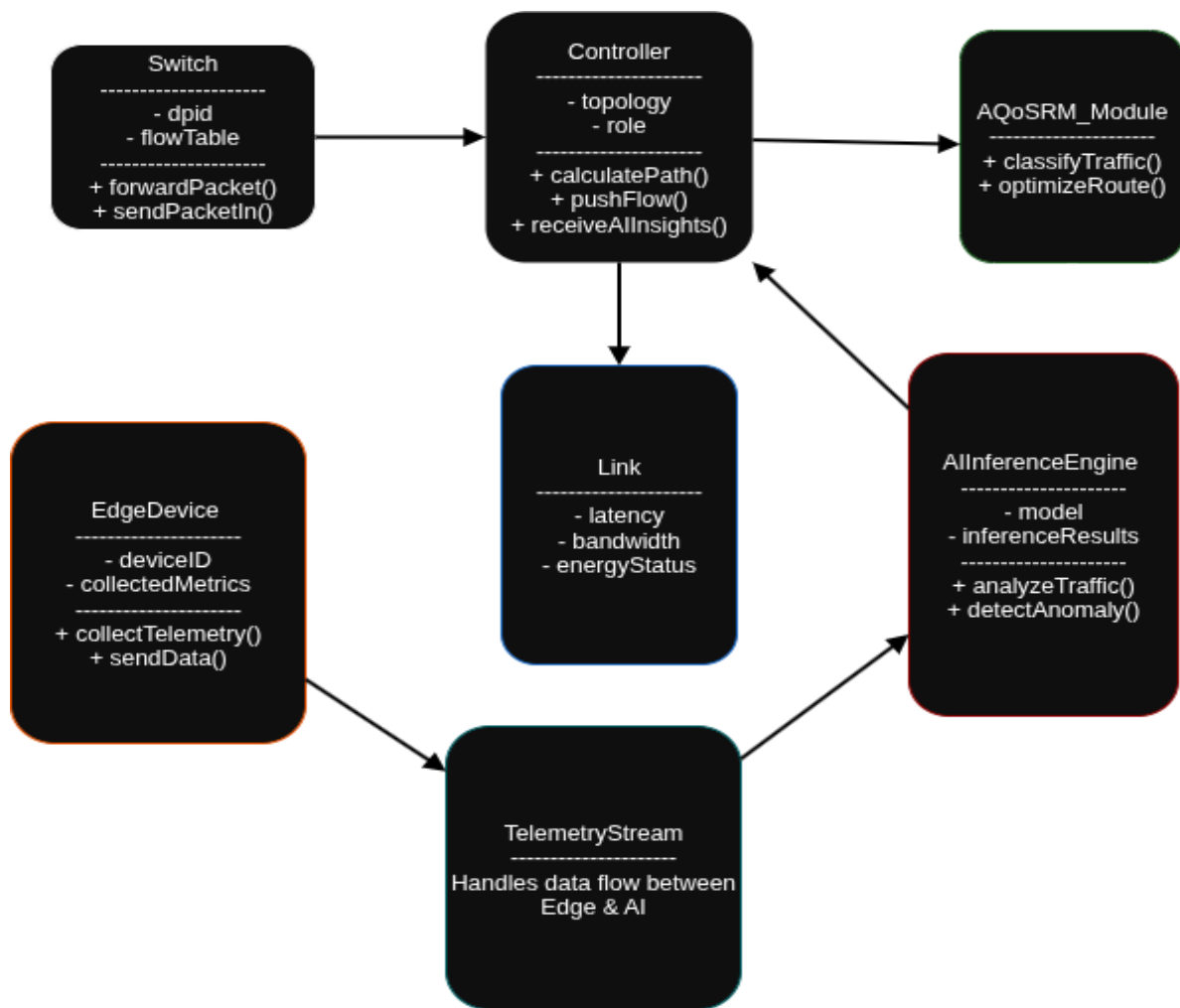
4.4.2 Non-Functional Requirements

- Scalability: Support network expansion
- Reliability: Maintain uptime under failures
- Security: Secure controller-switch communication
- Efficiency: Minimize control plane overload
- Cost-effectiveness: Use low-cost edge hardware

4.5 System Models: UML Diagrams

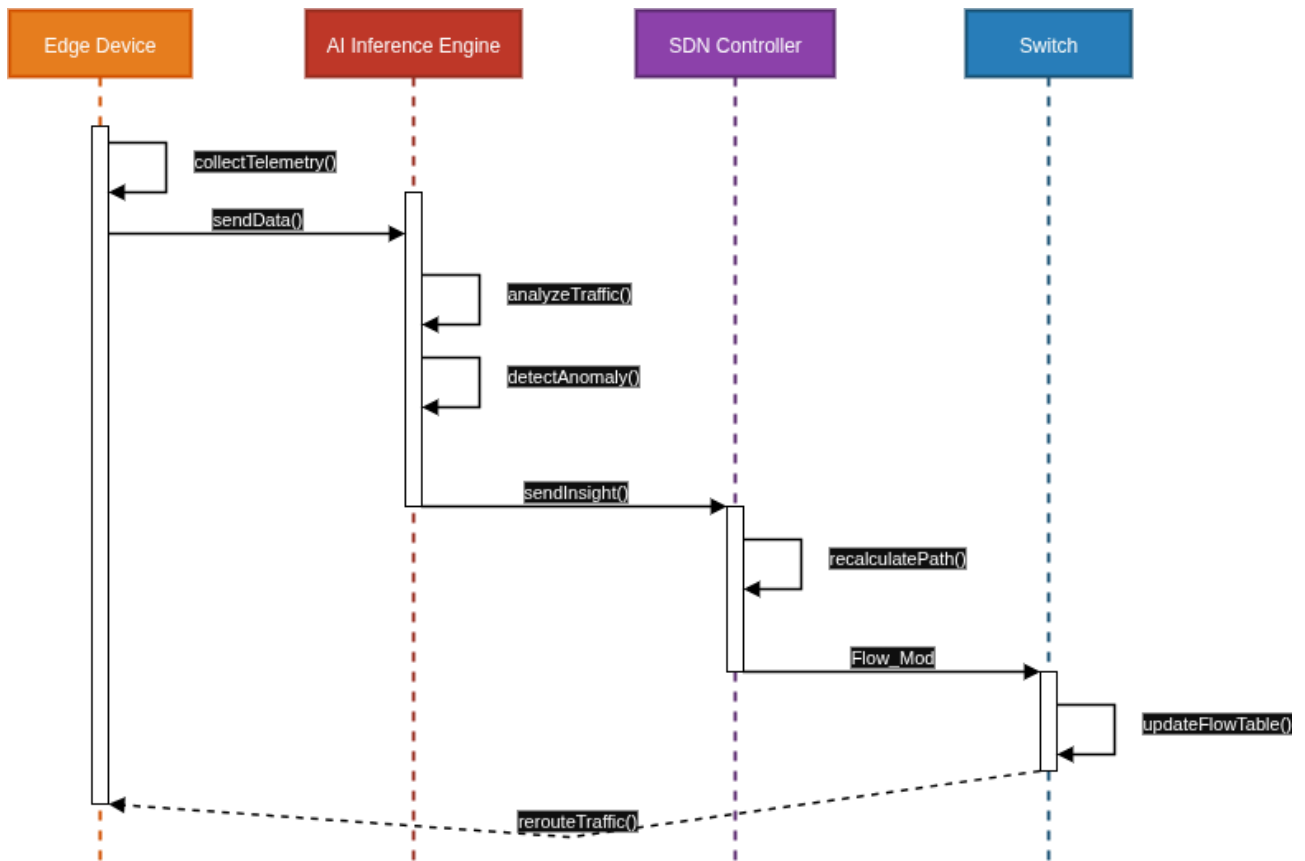
4.5.1 Class Diagram

Fig 2.1



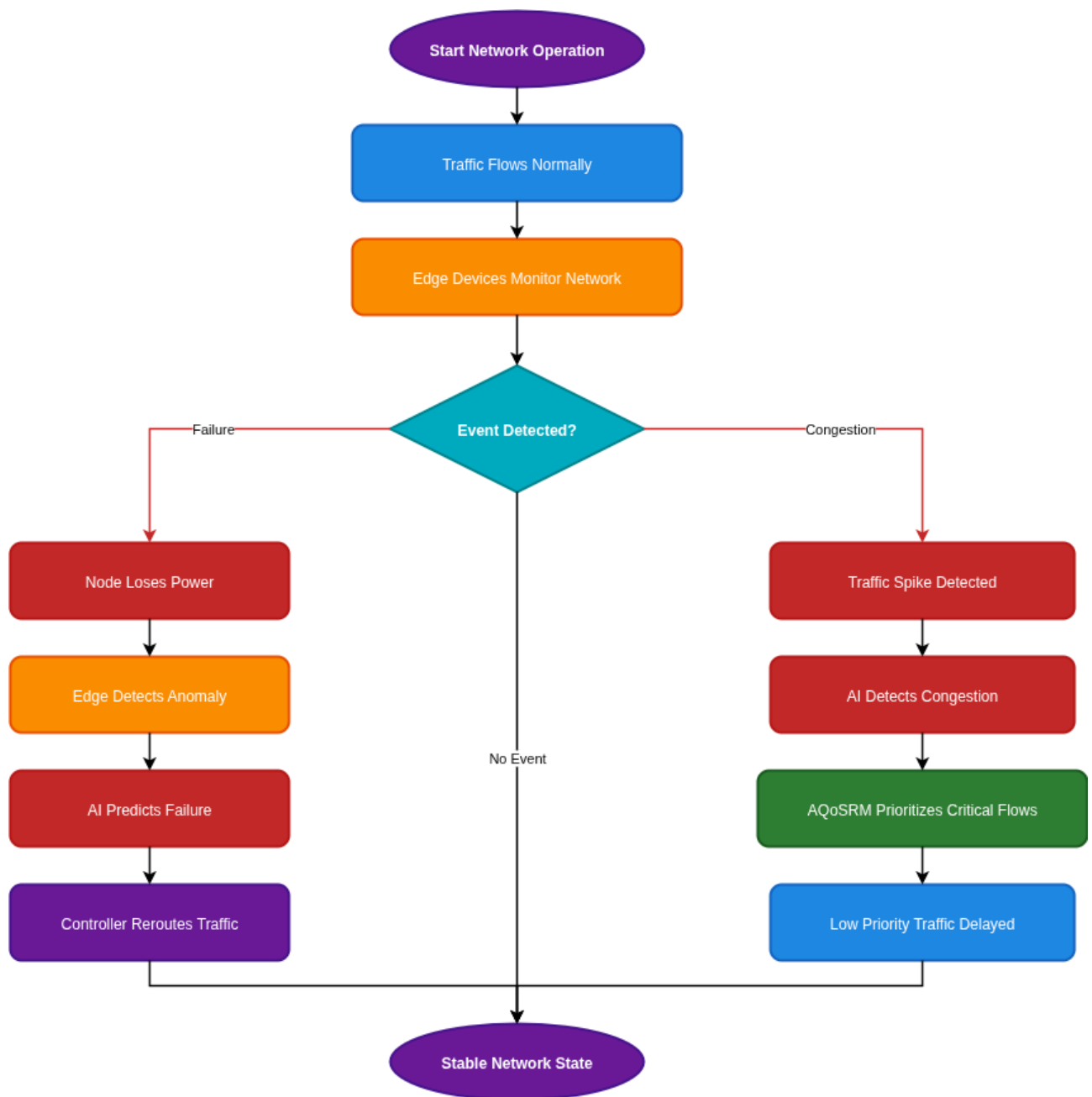
4.5.2 Sequence Diagram

Fig 2.2



4.6 System Workflow

Fig 2.3



4.7 Comparative Analysis

Table 2.0

Feature	Legacy System	ZAN Framework
Control	Distributed	Centralized and Intelligent
Intelligence	None	AI-driven

Failover	Slow	< 50 ms
QoS	Limited	Adaptive
Cost	High	Low (Hybrid)

4.8 Conclusion

The chapter came in with the analysis of the current network system and also the capabilities of the proposed ZAN framework, demonstrating that while legacy systems lack adaptability and resilience, the ZAN framework has effectively integrated the AI and Software-Defined Networking. The analysis confirms that the addition or integration of AI in networking processes enhances the real-time decision-making, allowing the system to act or respond dynamically to failures and network congestion. The proposed architecture ultimately provides a scalable, intelligent and cost-effective solution tailored to Zimbabwe's unique operational challenges.

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